Tiny Engineer - laptop screen inserts: Printed background

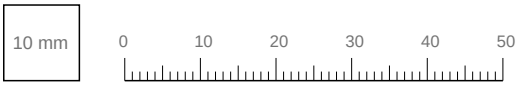
Servo preset: Tower Pro SG90 (sg90)

Print one page that suits your printer.

Background and text in colour; a black-only printer prints them as greys. Most ink.

Insert 45.6 x 34.4 mm (cut on the grey line). Visible window 45.2 x 25.9 mm, 0.2 mm from the left and 1.7 mm from the top edge.

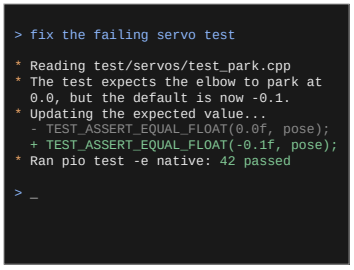
Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.



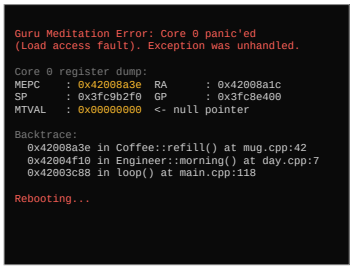
Square must measure 10 x 10 mm and the ruler 50 mm.
If not, reprint at 100% / Actual size (scaling off).

sg90 - Build output

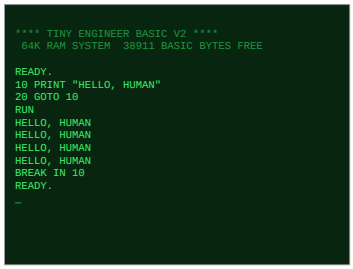
sg90 - Source code



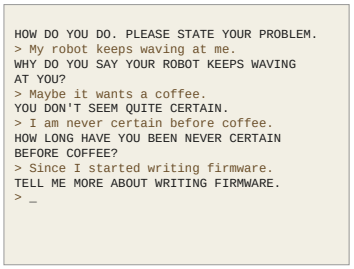
sg90 - AI agent chat



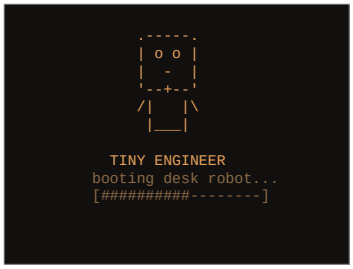
sg90 - Crash / stack trace



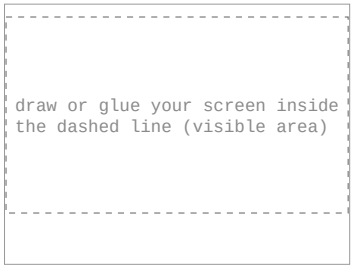
sg90 - Retro computer



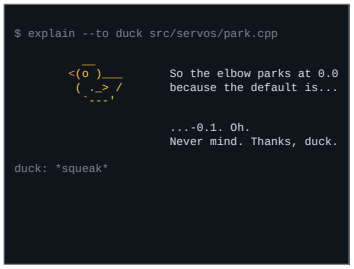
sg90 - ELIZA-style chat



sg90 - Tiny Engineer splash



sg90 - Blank template



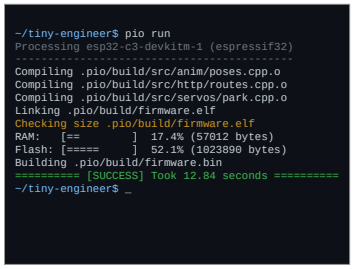
sg90 - Rubber duck debugging

sg90 - Linux boot (Tux)

sg90 - Pixel aliens

sg90 - Blocky neighbour

sg90 - TE-9000 console



sg90 - Build output

sg90 - Source code

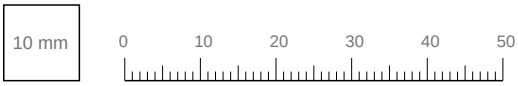
Tiny Engineer - laptop screen inserts: No background, colour text

Servo preset: Tower Pro SG90 (sg90)

Print one page that suits your printer.

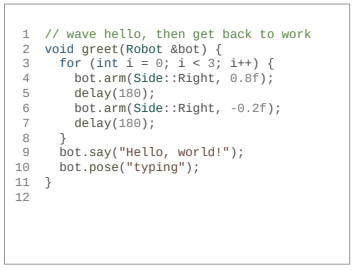
Colour printer. Only the text prints, darkened to read on pale paper; the paper shows through.

Insert 45.6 x 34.4 mm (cut on the grey line). Visible window 45.2 x 25.9 mm, 0.2 mm from the left and 1.7 mm from the top edge.
Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.

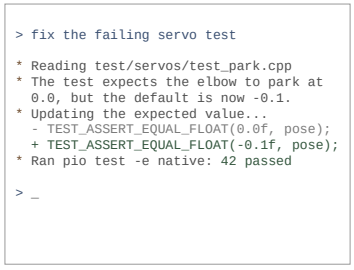


Square must measure 10 x 10 mm and the ruler 50 mm.
If not, reprint at 100% / Actual size (scaling off).

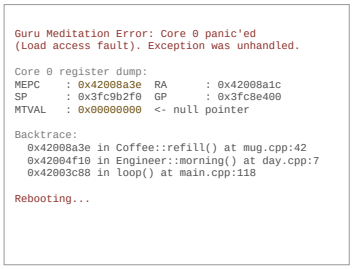
sg90 - Build output



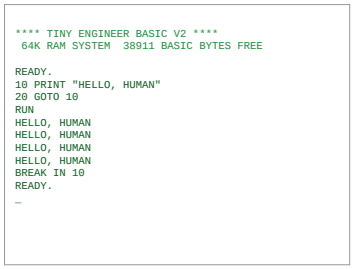
sg90 - Source code



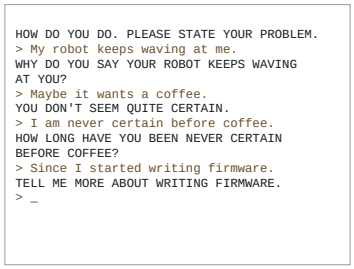
sg90 - AI agent chat



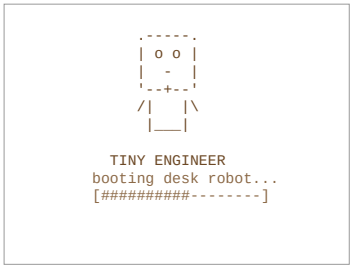
sg90 - Crash / stack trace



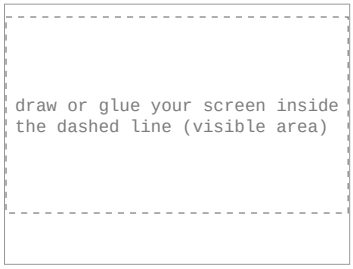
sg90 - Retro computer



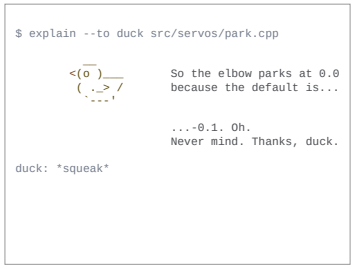
sg90 - ELIZA-style chat



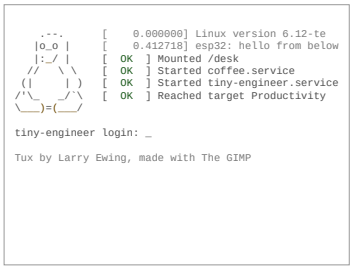
sg90 - Tiny Engineer splash



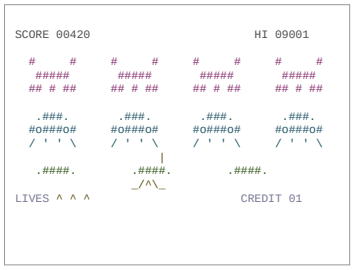
sg90 - Blank template



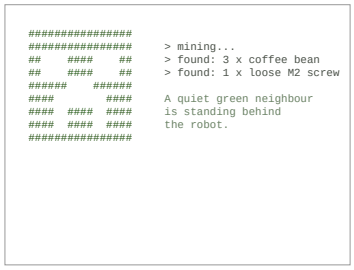
sg90 - Rubber duck debugging



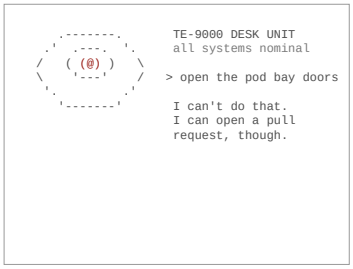
sg90 - Linux boot (Tux)



sg90 - Pixel aliens

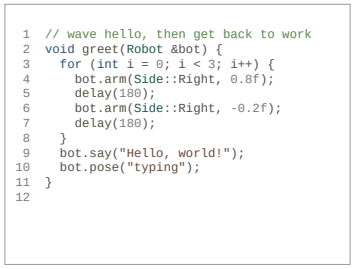


sg90 - Blocky neighbour



sg90 - TE-9000 console

sg90 - Build output



sg90 - Source code

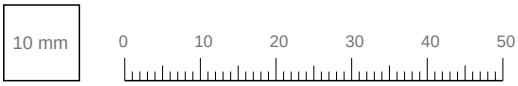
Tiny Engineer - laptop screen inserts: No background, black text

Servo preset: Tower Pro SG90 (sg90)

Print one page that suits your printer.

Any printer. Only the text prints, in black and grey; the paper shows through. Least ink.

Insert 45.6 x 34.4 mm (cut on the grey line). Visible window 45.2 x 25.9 mm, 0.2 mm from the left and 1.7 mm from the top edge.
Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.



Square must measure 10 x 10 mm and the ruler 50 mm.
If not, reprint at 100% / Actual size (scaling off).

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
--------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

sg90 - Build output

```
 1  // wave hello, then get back to work
 2  void greet(Robot &bot) {
 3    for (int i = 0; i < 3; i++) {
 4      bot.arm(Side::Right, 0.8f);
 5      delay(180);
 6      bot.arm(Side::Right, -0.2f);
 7      delay(180);
 8    }
 9    bot.say("Hello, world!");
10    bot.pose("typing");
11  }
12
```

sg90 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

sg90 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

sg90 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

sg90 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

sg90 - ELIZA-style chat

```
              .------.
              | o o |
              |  -   |
              '--+--'
             /|    |\
              |____|

          TINY ENGINEER
        booting desk robot...
        [###########--------]
```

sg90 - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

sg90 - Blank template

```
$ explain --to duck src/servos/park.cpp

        __
       <(o )___        So the elbow parks at 0.0
        ( ._> /        because the default is...
         `---'

                       ...-0.1. Oh.
                       Never mind. Thanks, duck.

duck: *squeak*
```

sg90 - Rubber duck debugging

```
   .--.      [    0.000000] Linux version 6.12-te
  |o_o |     [    0.412718] esp32: hello from below
  |:_/ |     [  OK  ] Mounted /desk
 //   \ \    [  OK  ] Started coffee.service
(|     | )   [  OK  ] Started tiny-engineer.service
/'\_   _/`\  [  OK  ] Reached target Productivity
\___)=(___/

tiny-engineer login: _

Tux by Larry Ewing, made with The GIMP
```

sg90 - Linux boot (Tux)

```
SCORE 00420                      HI 09001

  #     #     #     #     #     #     #
 #####       #####       #####       #####
## # ##     ## # ##     ## # ##     ## # ##

 .###.       .###.       .###.       .###.
#o###o#     #o###o#     #o###o#     #o###o#
/ ' ' \     / ' ' \     / ' ' \     / ' ' \
                 |
  .####.         .####.         .####.
                _/^\_
LIVES ^ ^ ^                    CREDIT 01
```

sg90 - Pixel aliens

```
################
################     > mining...
##    ####    ##     > found: 3 x coffee bean
##    ####    ##     > found: 1 x loose M2 screw
######    ######
####        ####     A quiet green neighbour
####  ####  ####     is standing behind
####  ####  ####     the robot.
################
```

sg90 - Blocky neighbour

```
    .-------.       TE-9000 DESK UNIT
  .'  .---.  '.     all systems nominal
 /   ( (@) )   \
 \    '---'    /   > open the pod bay doors
  '.         .'
    '-------'       I can't do that.
                    I can open a pull
                    request, though.
```

sg90 - TE-9000 console

```
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Processing esp32-c3-devkitm-1 (espressif32)
--------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
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sg90 - Build output

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sg90 - Source code